

SEDAR 63 Gulf Menhaden

Data and Assessment Workshops

Throughout the course of the Data Workshop and Assessment Workshop, a number of items were identified as important research topics for future stock assessments. The assessment panel evaluated the various items and developed a consensus priority list.

DATA ELEMENT	RECOMMENDATION	Priority
Genetics And Stock Structure	Improve species identifications at the periphery of the Gulf menhaden's range in Texas and Alabama/Florida waters for juveniles and adults.	High
Tagging Study (New)	Conduct Gulf menhaden tag/recovery study for better estimates of natural mortality, migration, growth, etc. which are inputs for the stock assessment.	High
Fishery-Independent Juvenile Index	Design and implement a survey dedicated to determining menhaden recruitment in the coastal rivers and upper bays of the northern Gulf of Mexico.	Med/High
Genetics And Stock Structure	Identify menhaden-specific nuclear DNA markers (preferably microsatellites or SNPs) using lab-based DNA library screening techniques. Evaluate these markers for use in genetic studies of Gulf menhaden.	Med/High
Fishery-Independent Adult Index	Collect and age Gulf menhaden scales from fishery-independent gears (e.g., gill nets) to determine selectivity. Expand efforts to age menhaden by state agencies.	Med
Stock Status Benchmarks	Research effort should be focused on determining appropriate reference points for the stock to ensure long term sustainability while balancing the desires of stakeholders to effectively exploit the stock.	Med
Modeling	Benchmarks – Develop procedures to establish assessment benchmarks (e.g., F or proxies) that account for the multiple priorities of ecosystem management that could include predation mortality and ecological yield separate from other forms of natural mortality.	Med
Recruitment Evaluation	Understanding the recruitment drivers for Gulf Menhaden that includes a number of environmental parameters.	Med/Low
Environmental Indices	Develop a habitat index to examine the potential shift in the Gulf menhaden population to more inshore waters as marsh converts to open water from coastal land loss.	Med/Low

Legacy Data (Fishery-Dependent Surveys)	Process and analyze samples that address the homogeneity of the catch in the hold of the reduction fishery vessels.	Med/Low
Predator/Prey	Expand understanding of diets of potential Gulf Menhaden predators using a variety of tools including traditional stomach analysis, DNA barcoding, and fatty acid profiles Gulf wide.	Med/Low
Legacy Data (Tagging Study)	Evaluate using current methods the historic archived tag data from Ahrenholz's original work.	Low
DATA ELEMENT	RECOMMENDATION	Priority
Fishery-Independent Adult Survey	Develop and implement an acoustic survey for menhaden populations during winter months to determine spatial distribution and abundance.	Low
Mortality Study	Evaluation of menhaden involved in 'fish kills' which can include impingement, red tides, freezes, and jubilees. Quantifying the additional (non-natural) mortality on populations.	Low
Modeling	Conduct additional research into simulation models such as MSVPAs, ECO-SIM, EcoPath, etc.; results could produce better estimates of natural mortality as well as other fishery parameters.	Low

Review Workshop

The Review Panel discussed the research recommendation from the Assessment Panel in detail. As stated above the main sources of uncertainty including 1) the sensitivity of the model to inclusion of the adult gillnet survey and associated length structure, 2) natural mortality, 3) lack of reference points or management benchmarks. Considering the main sources of uncertainty in this assessment, the Review Panel re-prioritized the Assessment Panels list and made a few additions, as outlined below.

Table 1. Research recommendations reviewed or offered by the Review Panel, including the Panel's indication of priority for implementation prior to the next Gulf menhaden assessment.

DATA ELEMENT	RECOMMENDATION	Priority
Tagging Analysis	Reevaluation of tag based natural mortality estimates including evaluation of tag data from Ahrenholz's original work	High
Stock Status Benchmarks	Use simulations of potential stock recruitment relationships, coupled with MSE, to examine single species reference points or management approaches	High

Modeling	Further explore alternative models, particularly ASPIC, and compare with the current model.	High
Modeling	Exploring finer time resolution (e.g. quarterly) on the model	High
Ageing	Continue to explore the effects of ageing error and ageing bias in the model	High
Ageing	In cooperation with state agencies, implement aging of fish caught in independent sampling to allow for use of ages in modeling	High
Genetics and Stock Structure	Improve species identifications at the periphery of the Gulf menhaden's range in Texas and Alabama/Florida waters for juveniles and adults.	Med
Tagging Study	Conduct Gulf menhaden tag/recovery study for better estimates of natural mortality, migration, growth, etc. which are inputs for the stock assessment. After achieve data analyzed	Med
Modeling	Explore further diagnostics and presentation of model uncertainty	Med
Predator/Prey	Expand understanding of diets of potential Gulf Menhaden predators using a variety of tools including traditional stomach analysis, DNA barcoding, and fatty acid profiles Gulf wide.	Med
Fishery-Independent Adult Index	Collect and age Gulf menhaden scales from fishery-independent gears (e.g., gill nets) to determine selectivity. Expand efforts to age menhaden by state agencies.	Med
Predator/Prey	Expand understanding of diets of potential Gulf Menhaden predators using a variety of tools including traditional stomach analysis, DNA barcoding, and fatty acid profiles Gulf wide.	Med
Fishery-Independent Juvenile Index	Design and implement a survey dedicated to determining menhaden recruitment in the coastal rivers and upper bays of the northern Gulf of Mexico.	Low
Genetics and Stock Structure	Identify menhaden-specific nuclear DNA markers (preferably microsatellites or SNPs) using lab-based DNA library screening techniques. Evaluate these markers for use in genetic studies of Gulf menhaden.	Low
Legacy Data (Fishery-Dependent Surveys)	Process and analyze samples that address the homogeneity of the catch in the hold of the reduction fishery vessels.	Low

While the Panel did not make detailed recommendations as to what should be improved in the SEDAR process, it is likely that the individual CIE review reports may contain some information in that respect. The Panel did agree that the recommendation to improve model diagnostics and presentation would be useful for this assessment, and a standardization presentation of that type of information would like improve the review of other SEDAR assessments.